

Sheridan

Linear Algebra
MATH15325D

Lecture 2 – Worksheet



Solving two equations

Solve each of the following two equations:

$$\begin{aligned}x + 3y &= 3 \\ 2x + 4y &= 9\end{aligned}$$

$$\begin{aligned}x + 3y &= 3 \\ 2x - 3y &= 9\end{aligned}$$

$$\begin{aligned}x + 3y &= 3 \\ 2x - 4y &= 7\end{aligned}$$

$$\begin{aligned}3x + 3y &= 5 \\ 2x - 4y &= 7\end{aligned}$$

Row operation

Determine which of the given answers is the solution.

$$2x_1 - 4x_2 - x_3 = 1$$

$$x_1 - 3x_2 + x_3 = 1$$

$$3x_1 - 5x_2 - 3x_3 = 1$$

a. $(3, 1, 1)$

b. $(3, -1, 1)$

c. $(13, 5, 2)$

d. $\left(\frac{13}{2}, \frac{5}{2}, 2\right)$

e. $(17, 7, 5)$

Row operation

Solve the following system of equations:

$$3x + 6y - 5z = 0$$

$$2x + 4y - 3z = 1$$

$$x + y + 2z = 9$$

Row operation

Solve the following system of equations:

$$3x - 3y + 6z = 15$$

$$2x - 2y + 4z = 10$$

$$x - y + 2z = 5$$

Back-substitution

The following augmented matrix was achieved by row-operation for a system of equations. Find the values of the variables x_1 , x_2 , x_3 , x_4

$$\left[\begin{array}{cccc|c} 1 & -3 & 4 & 2 & 5 \\ 0 & 1 & 2 & 4 & 1 \\ 0 & 0 & 1 & 6 & 9 \\ 0 & 0 & 0 & 1 & 2 \end{array} \right]$$

Gaussian Elimination

Solve by Gauss-elimination.

$$x_1 + 3x_2 - 2x_3 + 2x_5 = 0$$

$$2x_1 + 6x_2 - 5x_3 - 2x_4 + 4x_5 = -1$$

$$5x_3 + 10x_4 + 15x_5 = 5$$

$$2x_1 + 6x_2 + 8x_4 + 4x_5 + 18x_6 = 6$$